

Three distinct voters in a one-dimensional single-peaked, single-crossing profile Disproof of conjecture 00000008244

ziangni-sys

4 October 2026

The source's middle conjunct states that the maximal agent count in the simultaneous single-peaked and single-crossing domain is $2^d d$. In dimension one this is two. An actual one-dimensional Euclidean profile with three distinct strict preferences disproves that bound. We use neither duplicate preferences nor a higher-dimensional embedding.

Actual locations and utilities. There are three alternatives a, b, c with locations

$$q_a = 0, \quad q_b = 2, \quad q_c = 3$$

on the real line. Three agents have distinct locations

$$p_0 = 0, \quad p_1 = 5/4, \quad p_2 = 7/4.$$

The actual Euclidean utility of agent i for alternative z is

$$u_i(z) = -(q_z - p_i)^2.$$

These give the complete table

i	$u_i(a)$	$u_i(b)$	$u_i(c)$	strict order
0	0	-4	-9	$a \succ b \succ c$
1	-25/16	-9/16	-49/16	$b \succ a \succ c$
2	-49/16	-1/16	-25/16	$b \succ c \succ a$

Every agent has a strict total order, with no ties, and all three orders are distinct. The alternative axis is $a < b < c$; the voter order is $0 < 1 < 2$, agreeing with their actual real locations.

Single-peakedness. Agent zero has peak a and prefers alternatives less as one moves right. Agents one and two have peak b , with b preferred to the sole alternative on each side. Equivalently, each preference increases toward its peak along the fixed axis and decreases away from it on either side. Thus the complete profile is single-peaked on this one-dimensional axis.

Single-crossing, including all alternative pairs. Along the voter order $0 < 1 < 2$, the strict-comparison sets are

$$\{i : a \succ_i b\} = \{0\}, \quad \{i : a \succ_i c\} = \{0, 1\}, \quad \{i : b \succ_i c\} = \{0, 1, 2\}.$$

Each is an initial segment. Each reversed comparison is its complement, a final segment; a comparison of an alternative to itself is empty. Hence every ordered-pair comparison set is an initial or final segment of the voter order. This is the standard single-crossing condition: a pairwise preference can switch at most once. In particular a versus b switches after voter zero, a versus c after voter one, and b versus c never switches.

The actual dimension and failure of the count bound. Both agents and alternatives are located in $\mathbb{R}$ as a real vector space. Its actual real dimension is one:

$$d = \dim_{\mathbb{R}} \mathbb{R} = 1.$$

There are three agents and three distinct preference orders in the simultaneous domain. Therefore

$$3 > 2 = 2^1 \cdot 1.$$

Even counting distinct preferences instead of agents does not rescue the claimed bound.

The profile also has the usual median-voter behavior: alternative b , the peak of the middle voter, defeats a by two votes to one and defeats c unanimously. Thus the counterexample does not depend on a failure of median existence.

Scope. This refutes the stated maximal-agent-count conjunct. It makes no claim about the other conjectural criteria or the separate size formula for the full single-peaked domain. Single-crossing is verified for an explicit voter order, and single-peakedness for the same fixed alternative axis.

Formal correspondence. Lean defines actual real locations and negative squared-distance utilities. An explicit finite rank table represents the three orders, and the utility/preference theorem proves equivalence to comparison of the actual utilities for every agent and pair.

The proof verifies strict total preferences and pairwise distinctness. **SinglePeaked** is the full increase-to-peak, decrease-from-peak condition. **SingleCrossing** requires every ordered-pair comparison set to be downward or upward closed. Both properties are proved on the actual finite profile. The actual `Module.finrank` of $\mathbb{R}$ is one, and the final theorem combines all profile properties with the strict failure of the displayed count bound.

Reproduction. Run `lake build in lean/` with Lean 4.19.0 and public Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b`

The project prints axiom audits.